Time-Series in Geoscience: Differencing & Durbin-Watson

GeoPython — Quick Reference Handout 06

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1 Autocorrelation: Why Geoscience Data Is "Sticky"

In environmental time-series, consecutive observations are rarely independent. Yesterday's sea ice extent strongly predicts today's. This is autocorrelation — the variable is correlated with its own past values.

> **!** *Autocorrelated residuals violate a core assumption of linear regression. If your residuals are not independent, your p-values and confidence intervals are unreliable — even if R^2 looks good.*

2 Differencing: Changing the Question

Instead of modelling the level of a variable (How much ice is there?), model the change (How much did the ice change today?). This is differencing, and it often resolves autocorrelation without changing the model class.

	Level Model	Differenced (Delta) Model
Target variable	Ice extent today (absolute)	Δ Ice extent = today − yesterday
Typical residual pattern	Curved, systematic drift	Closer to random noise
Interpretation	"How much ice is there?"	"How much did conditions change?"
When to use	Data is stationary (no trend)	Data has trend or autocorrelation

3 The Durbin-Watson Statistic

The Durbin-Watson (DW) test measures autocorrelation in model residuals. The statistic ranges from 0 to 4.

DW Value	Interpretation	Action
≈ 2.0	✓ No autocorrelation — residuals independent	Model is valid; proceed
0 – 1.5	⚠ Positive autocorrelation	Consider differencing or lag features
2.5 – 4.0	⚠ Negative autocorrelation	Check for over-differencing or model misspecification

> 👁 *Goal = 2.0. Use this as your concrete target when validating time-series models. A DW close to 2 is a strong signal that your residuals are behaving as independent noise.*

> **!** *Differencing does not always mean a "bigger" or "better" model — it means a better question. Changing the target variable can dramatically improve DW without any model change.*